BIPRO ROY

AI-Driven Learning Analyst | Machine Learning, NLP & Learning Analytics
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Professional Summary

Final-semester Computer Science and Engineering student at AIUB who builds AI-powered learning tools and the analytics behind them. Developed FluentUp, an English-learning app that places learners with an assessment, adapts their path, schedules review with spaced repetition and tracks their progress, with a Gemini-based AI tutor. Experienced in fine-tuning Transformer models for Bangla NLP and in comparing machine learning and statistical forecasting models, with a first-author paper presented at ICIS 2025. Looking to use machine learning, NLP and data analysis to measure and improve how people learn.

Technical Skills

- **Programming:** Python, TypeScript, JavaScript, SQL, C++
- **Machine Learning & NLP:** Scikit-learn, Hugging Face Transformers (DistilBERT fine-tuning), PyTorch, LSTM, SVR, ARIMA, Prophet, text preprocessing, regex-based information extraction, stemming, inverted indexing
- **LLM Integration:** Gemini API, prompting for structured JSON output, response validation with Pydantic, rule-based fallbacks
- **Data Analysis & Visualization:** Pandas, NumPy, Recharts dashboards, time-series model evaluation (RMSE, MAE, MAPE), classification metrics
- **Learning Analytics:** placement assessment, spaced repetition (SM-2), learner progress tracking (XP, streaks, lesson completion), mistake tracking
- **Development & Tools:** FastAPI, Streamlit, Next.js, React Native (Expo), PostgreSQL, SQLAlchemy, Docker, Git, GitHub Actions, pytest, Playwright

Education

B.Sc. in Computer Science and Engineering | American International University-Bangladesh (AIUB), Dhaka | 2023 – Expected April 2027

- CGPA: 3.40 / 4.00 (final semester)
- Relevant coursework: Machine Learning, Natural Language Processing, Data Structures and Algorithms, Database Management Systems, Web Technologies

Higher Secondary Certificate (Science) | Gaokhali School and College | 2022 | GPA 5.00 / 5.00

Relevant Projects

FluentUp – AI-Powered English Learning App | React Native, TypeScript, FastAPI, PostgreSQL, Gemini API | github.com/roybipro/fluentup

- Built an English-learning app for Bengali-speaking learners in which a weighted 5-question placement assessment recommends a starting CEFR or IELTS learning path.
- Designed a learning engine with 6 levelled paths, lesson prerequisites, 6 quiz types and a mistake bank that stores each learner's errors for targeted review.
- Implemented the SM-2 spaced-repetition algorithm to schedule vocabulary review from each learner's recall performance.
- Built a progress dashboard that tracks XP, current and longest streaks, completed lessons and achievements.
- Integrated the Gemini API on the server for an AI conversation tutor, grammar feedback and IELTS answer evaluation, validating every response with Pydantic and falling back to rule-based checks when the API is unavailable.
- Wrote 60 backend (pytest) and 18 mobile (Jest) tests; secured accounts with Supabase Auth and PostgreSQL row-level security.

Bangla Fake News Detector | Python, Hugging Face Transformers, PyTorch, Streamlit | github.com/roybipro/fake-news-project

- Fine-tuned multilingual DistilBERT to classify Bangla news articles as authentic or fake from combined headline and body text.
- Handled class imbalance (about 7,000 authentic vs. 1,000 fake articles) by undersampling to a balanced training set.
- Evaluated on a held-out test set with accuracy, precision, recall and F1.
- Deployed an interactive Streamlit app for single-article predictions.

CareerLab – Career and Research Tracker | FastAPI, PostgreSQL, Next.js, TypeScript | Live: careerlab-opal.vercel.app

- Built a full-stack workspace for tracking job applications, research opportunities and research projects, with Kanban boards, deadlines and reminders.
- Created an analytics page with status charts and monthly application trends computed from the database.
- Covered the system with 239 backend tests against a real PostgreSQL database and Playwright end-to-end tests.

AIUB Faculty Data Collector (NLP coursework) | Python, regular expressions | github.com/roybipro/NLP-Faculty-Data-Scraper-AIUB

- Extracted names, e-mails and room numbers for 197 faculty members using regular expressions only, after locating the JSON feed behind a JavaScript-rendered page.
- Built stemming, an inverted index of 641 stems and a regex query tool over faculty research-interest text.

Research Publications

- **Roy, B.** "AI for Stock Market Forecasting in Emerging Economies." 7th International Conference on Integrated Sciences (ICIS 2025), Dhaka, Bangladesh, 25 October 2025. Presented. Supervisor: Md. Mortuza Ahmmed. aiub.edu/Files/student-research/AI_for_Stock_Market_Forecasting_in_Emerging_Economies.html

Research Experience

Undergraduate Researcher | American International University-Bangladesh | 2025

- Compared LSTM, Support Vector Regression, Prophet and ARIMA for forecasting Bangladeshi bank share prices using Dhaka Stock Exchange, Yahoo Finance and Kaggle data.
- Evaluated models with RMSE, MAE and MAPE; ARIMA achieved the lowest test error (RMSE 0.91, MAE 0.64, MAPE 1.17%).
- Presented the findings at ICIS 2025 under the supervision of Md. Mortuza Ahmmed.

Research interests: Natural Language Processing, Bangla NLP, Large Language Models, LLM reliability and hallucination detection, AI in education

Languages

Bengali (native), English (professional working proficiency)